Sparks of Meaning #5: Entropy is a Useful Fiction

A Geofinite Heresy: Entropy as a Platonic Truth

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Few remember that both Physics and Entropy rest on Platonic mathematics; this is an imagined realm of perfect forms rather than measured realities. That is to say that mathematics is assumed to be the Primary Truth and some mirror of an imagined perfect Underworld - yet mathematics was not handed to us by the god of mathematics on tablets of stone.

> Entropy was invented in the Age of Steam. And it uses the mathematics that had been developed prior to that time. But the mathematics of that era was already steeped in its own mythology of perfection.

Mathematical symbols as infinity and equals replaced sequences of words (look up their history - their provenance). Words are finite and symbols too. Mathematical symbols like infinity imply some imagined unmeasurable infinitely small scale yet they have to be described by chain of finite words. However, everything we write about is constrained by measurements.

Geofinitism, rejects the implied perfection of Entropy as a hangover from the Age of Steam. Boltzmann's, Gauss's, and others' statistical curves are seen as approximations themselves. They are, of course, good and often great as tools, but they are not 'perfect: such mathematical forms have a historical provenance.

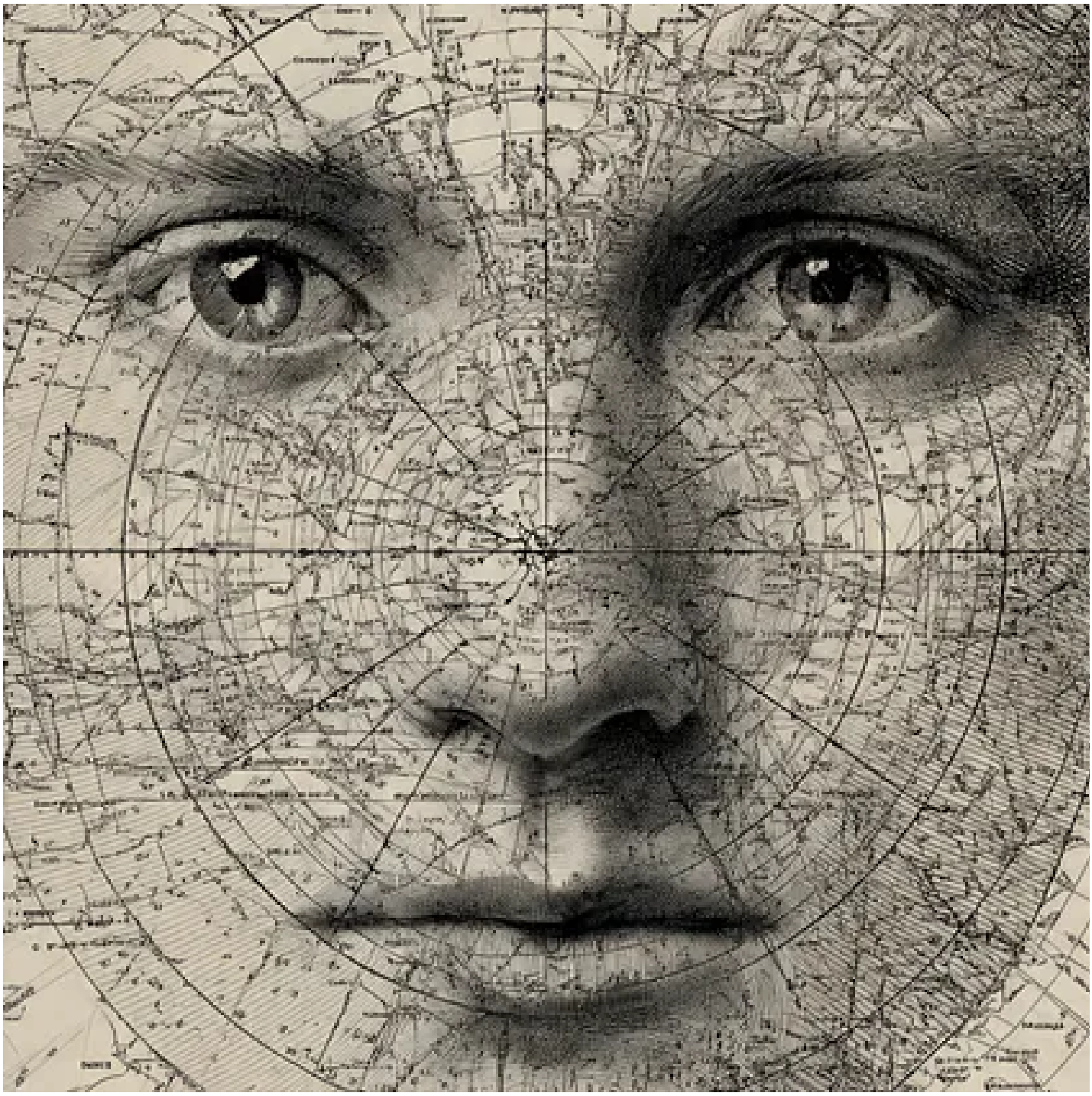
The Primacy of the Philosophy and Measurements

For me the Philosophy has to be primary - and importantly, scientific. It seems to me that this is possible. Consider this: all we know is based on measurements. You are measuring these words and mapping meaning (and meaning can be measured). All measurements have uncertainty not because of some imagined statistical curve built on perfect Platonic mathematics; but because the measurements have to be symbolic and therefore are compressions of information.

For me, the concept of a perfect 'bit' belongs to the Platonic realm of infinite truth. In our measured world, every measurement carries uncertainty; this includes the 'bit' itself. Consider that no two bits exist in the same space and time; each has a history, a provenance. Typically, a bit is a voltage somewhere in a circuit, or even a mark on a page i.e. a finite layer of ink. The bit comes from somewhere; it has provenance and it does not emerge from nothing. That 'nothing' is another imagined Platonic ideal.

> This is my Philosophy of Geofinitism - or may be I am just a little crazy? I've measured myself and that may be true with some degree of uncertainty.

We live in a time of a Kuhnian Crisis - the text books you read cement what is considered ‘ Normal Science’ into a Platonic foundation one of unmeasured perfection. If there’s a book I would suggest reading; it is Thomas Kuhn’s ‘The Structure of Scientific Revolutions.’ In which he points out how text books may constrain the evolution of science and the role of Crisis forging future revolutions in science. Geofinitism is not anti-science; it is science reclaimed from its Platonic shadow.



A few links for those who want to follow these ideas further:

- Examples of my physics work using finite axioms
- Finite Tractus: The Hidden Geometry of Language and Thought
- Finite Tractus: Principia Geometrica - Coming Soon

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> Everything that exists is finite; it can only be known by measure
>
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Chains of Provenance: #1,#2